Signal Transduction

"From the membrane to the nucleus and beyond…"

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The class meets on Monday from 2:00 – 5:30 p.m. in Lecture Room (Xian1-115)

September 7	**Lecture 1**	**An Overview**
	Lecture 2	**Receptors and Adaptors**
	Lecture 3	**G proteins, Ras, and other small GTP-binding proteins**
	Lecture 4	**Class discussion**
September 21	**Lecture 5**	**Post-translational Modification (PTM)**
	Lecture 6	**Protein Kinase and Phosphatase**
	Lecture 7	**c-Jun/AP-1 and JNK Signaling Pathway**
	Lecture 8	**NF-κB/IκBα and IKK Signaling Pathway**

Lecture 1 Signal Transduction - An Overview

What is signal transduction?

A process by which a chemical or physical signal is transmitted into a cell as a series of molecular events, most commonly protein phosphorylation, which ultimately results in a cellular response, such as proliferation, differentiation, programmed cell death, and tumorigenesis.

What is a signaling pathway?

“A chain of biochemical events elicited by binding of a signal to its receptor, leading to the changes in cell function and beyond…”

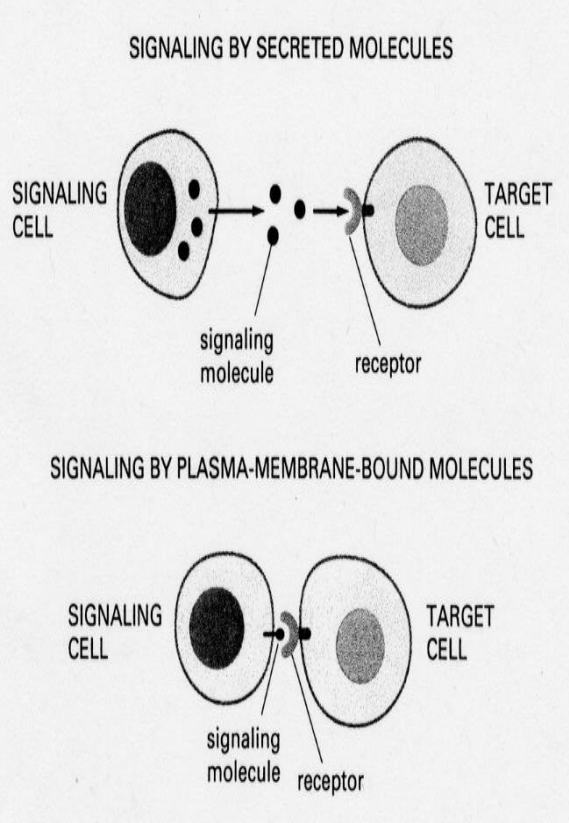
The General Structure of Signal Transduction

Signal	Growth factors, Cytokines, Proto-oncogenes, Stress, Tumor promoters, Virus and Hormones
↓	
Receptor	RTK, RS/TK, 7-TM
↓	
Adaptor	SH2/SH3, PH, PTB, PDZ and EH domain proteins
↓	
Amplifier/switcher	Protein kinases and GTP-binding protein
↓	
Effector	Transcription factors, cytoskeleton proteins
↓	
Genotype	Gene expression or suppression
↓	
Phenotype	Growth, differentiation, transformation and programmed death
↓	
Organism	Development, diseases, and cancer

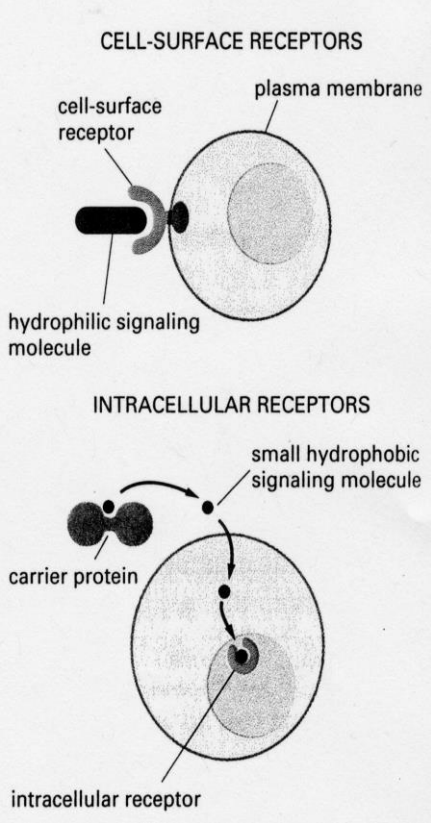
Note: NOT all signals are transmitted through all steps along the signaling cascade.

The Types of Cell Signaling

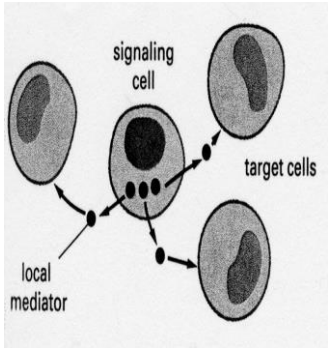
1. Inter-cellular signaling

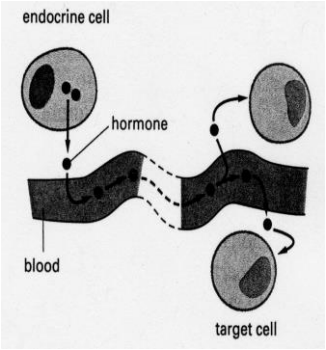


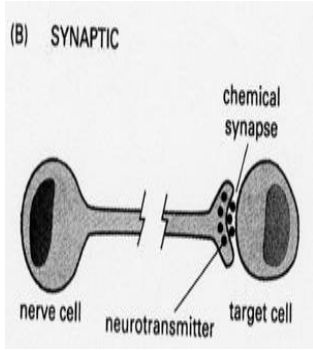
2. Extra-cellular signaling



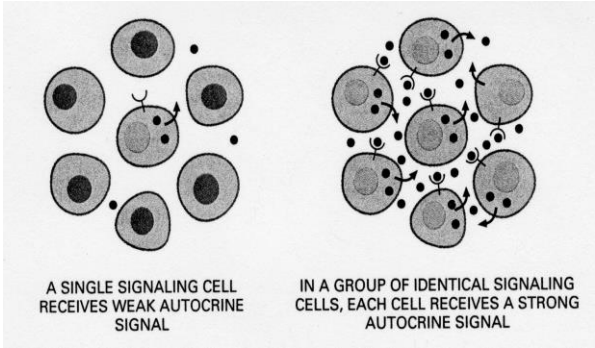
3. Paracrine



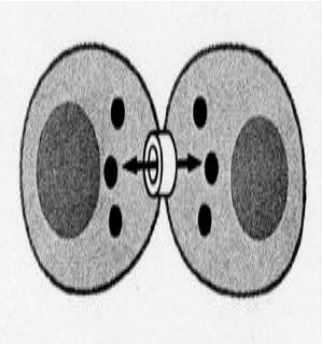




4. Autocrine



5. Gap junctions



Three-dimensional View of Signal Transduction

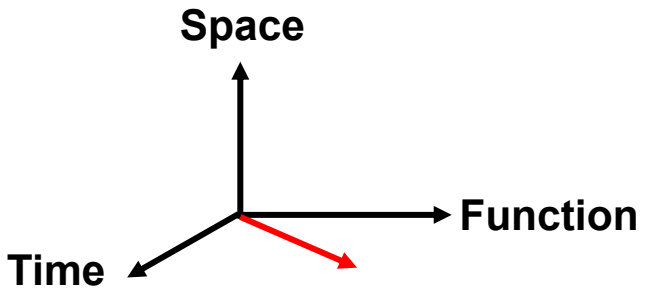
Temporal regulation (Time machine)

✓**Dynamics**

- Time course
- Pattern

✓**Visualization**

- Live cell image
- Green fluorescent protein (GFP)



Is there a 4th dimension?

Spatial regulation (Signaling domains)

✓**Membrane domains**

- Caveolae
- Focal adhesion
- PDZ domain

Three-dimensional View of Signal Transduction (cont'd)

Spatial regulation (Signaling domains) (cont'd)

✓Intracellular domains

- Anchoring proteins
- 14-3-3

✓Scaffold proteins

- Ste5
- JIP
- MP1

✓Subcellular compartment translocations

- NLS: a high proportion of the basic amino acid Lys and Arg;
- NES: Poorly defined motif;
- Phosphorylation-dependent: phosphorylation of PKM2 by ERK1/2
- Dimerization-dependent: ERK1/ERK2 nuclear translocation

Three-dimensional View of Signal Transduction (cont'd)

Functional regulation

✓Posttranslational modification (PTM)

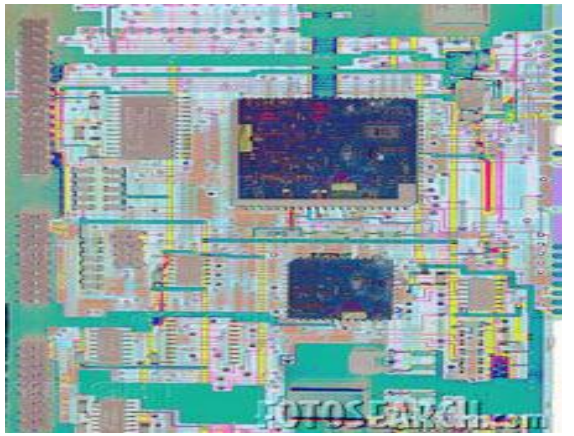
- Phosphorylation and dephosphorylation
- Ubiquitination-mediated activation and degradation

✓Epigenetic modification

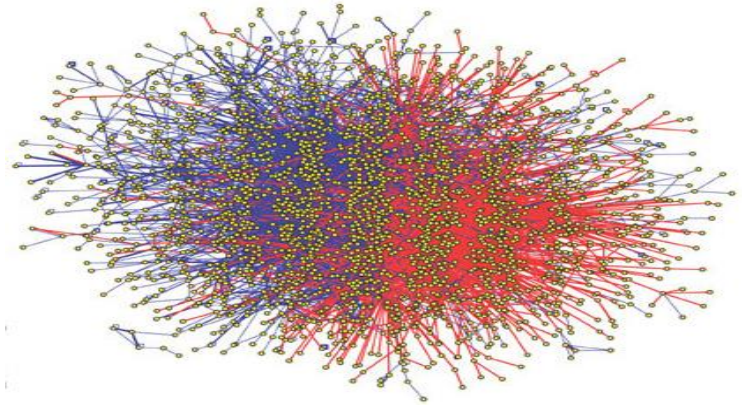
- Histone methylation
- Histone acetylation
- Histone ubiquitination
- Histone phosphorylation

✓Motifs and network integration

- Positive/negative feedback:
- Cross-talk
- Synergism/antagonism
- Signaling network wiring



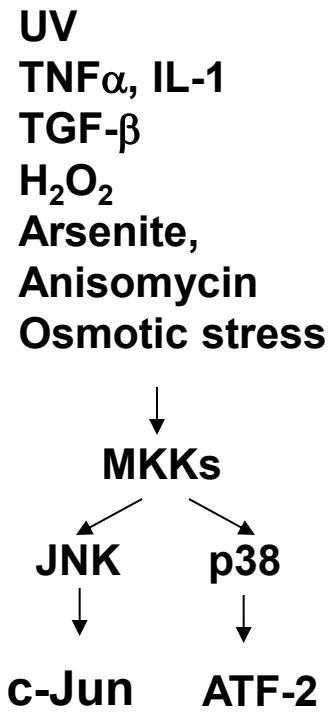
An electronic circuitry



A 3-dimensional cell signaling network

Regulation of Signal Transduction

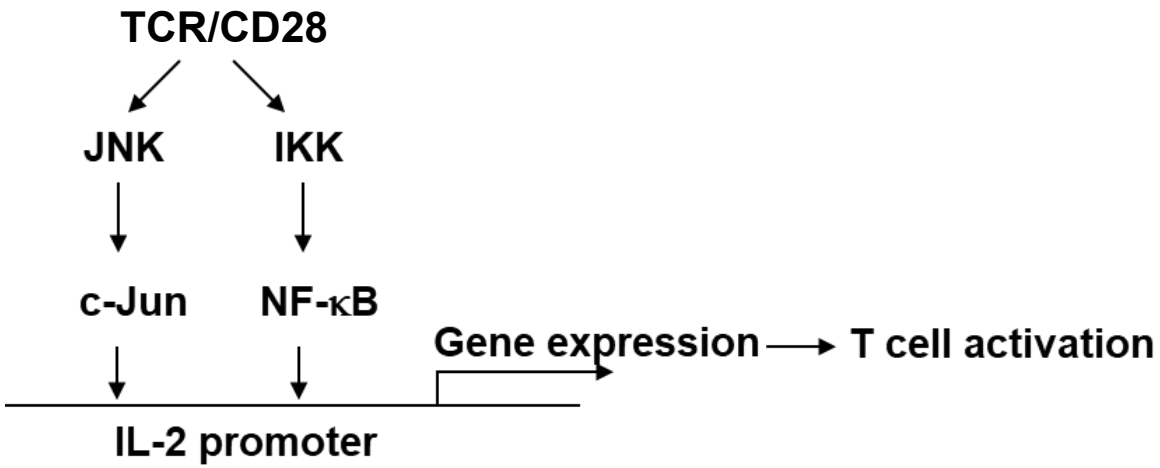
- ✓ **Signaling coordination**
  - ➢ **Coordinate activation of JNK and p38 pathways**



Regulation of Signal Transduction

✓ **Signaling coordination (cont'd)**

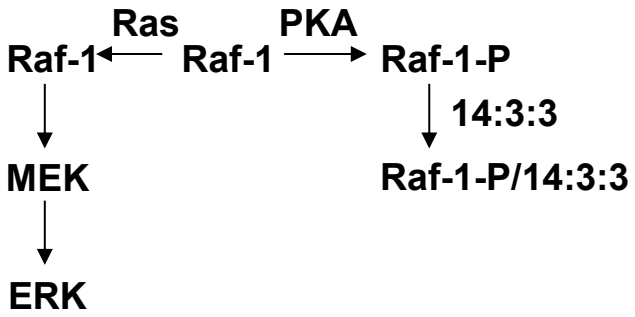
➢ **Transcriptional synergy between c-Jun and NF-κB**



Regulation of Signal Transduction (cont'd)

✓ Signaling negative regulation

➢ Negative regulation of the ERK pathway by cAMP



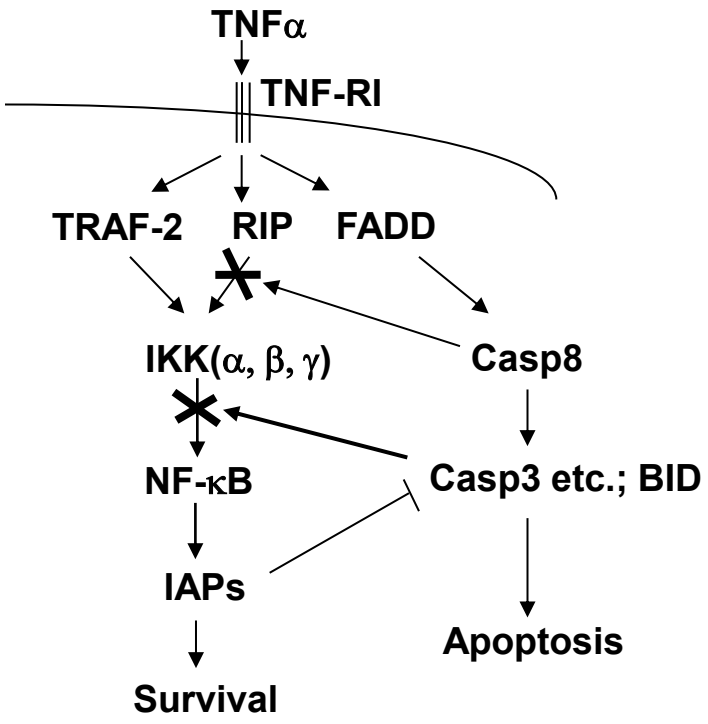
- Mechanism: Provides phospho-serines for 14:3:3 binding, which "locks" Raf-1 in an inactive conformation
- *In vivo*: S43
- *In vitro*: S43, S259 and S621
- Mutagenesis: S259 makes Raf refractory to PKA inhibition

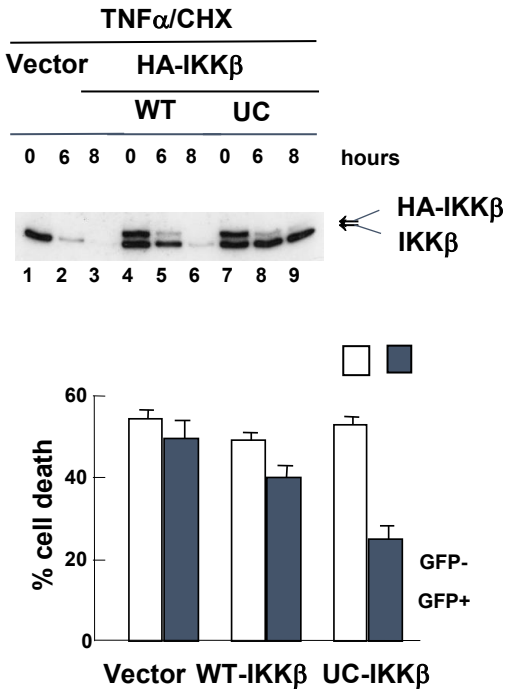
Wu et al., *Science 1993;*
Cook, S. J., and McCormick, F. *Science* 1993

Regulation of Signal Transduction (cont'd)

✓ Signaling negative regulation

➢ Negative regulation of the NF-κB pathway by caspases



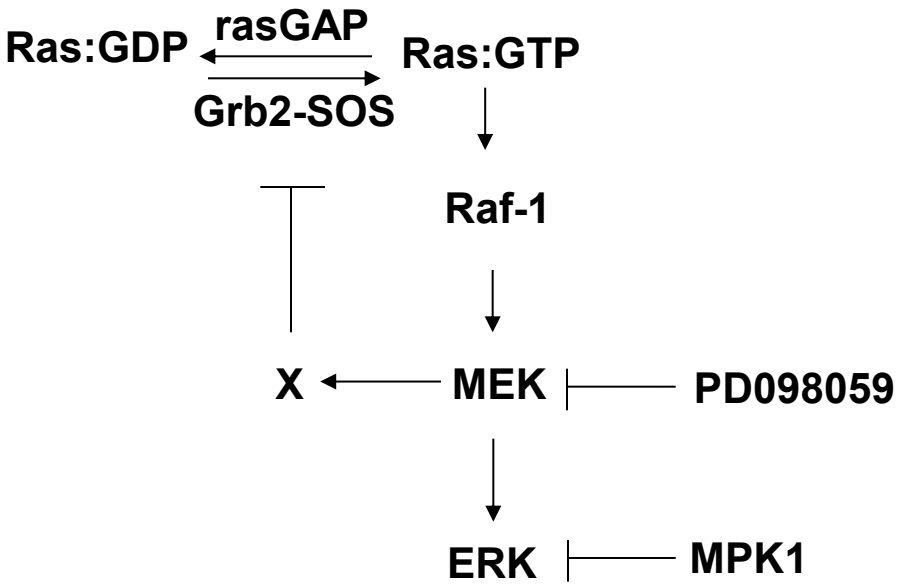


Tang *et al., Mol Cell* 2001

Regulation of Signal Transduction (cont'd)

✓ **Signaling feedback**

➢ **Inhibitory feedback by MEK on Ras signaling**

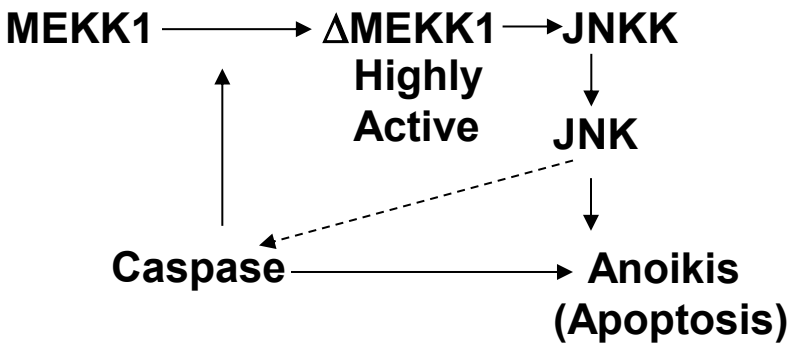


Holt *et al., MCB* 1996

Regulation of Signal Transduction (cont'd)

✓ Signaling feedback

➢ Stimulatory feedback by MEKK1 on caspase activation



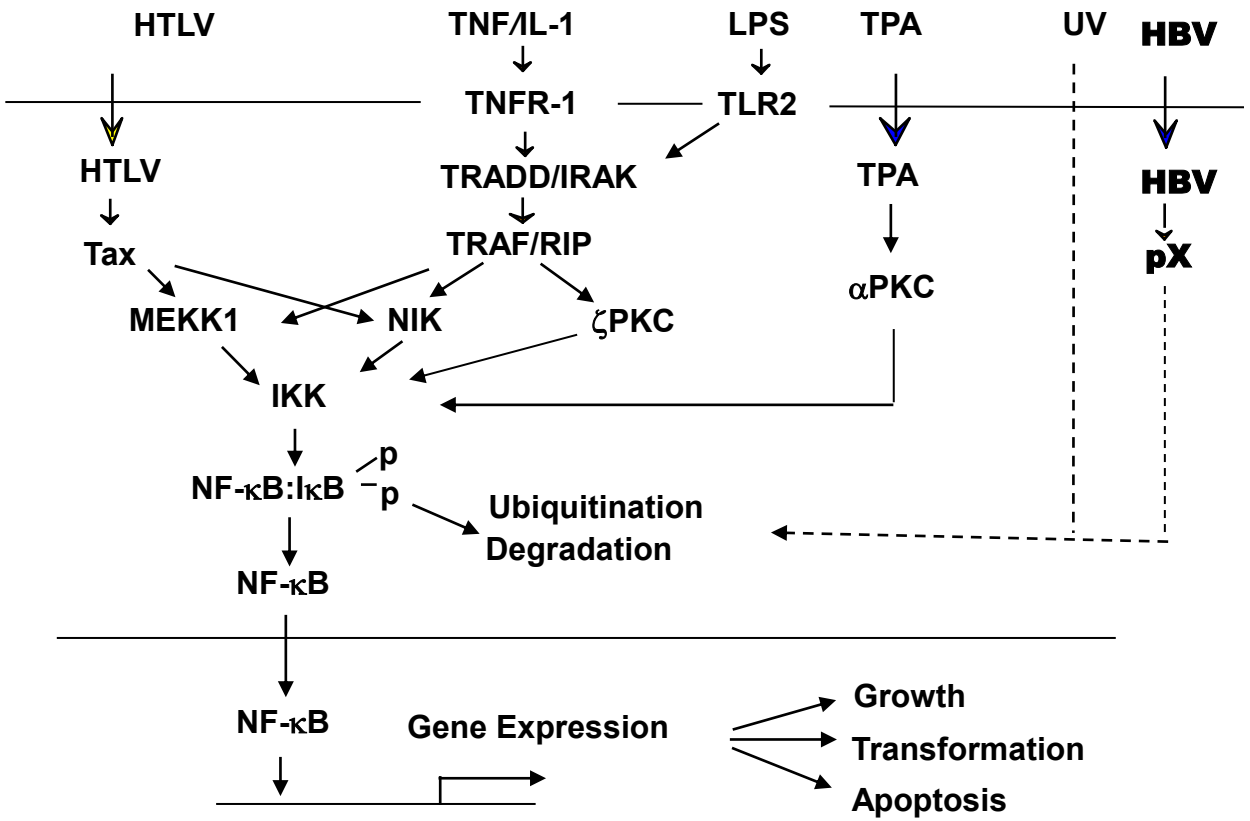
- Activation of MEKK1 requires cleavage by caspases
- Activation of JNK by MEKK1 may contribute to apoptosis through promoting cytochrome C release.

Cardone *et al., Cell* 1997

Regulation of Signal Transduction (cont'd)

✓ Signaling convergency

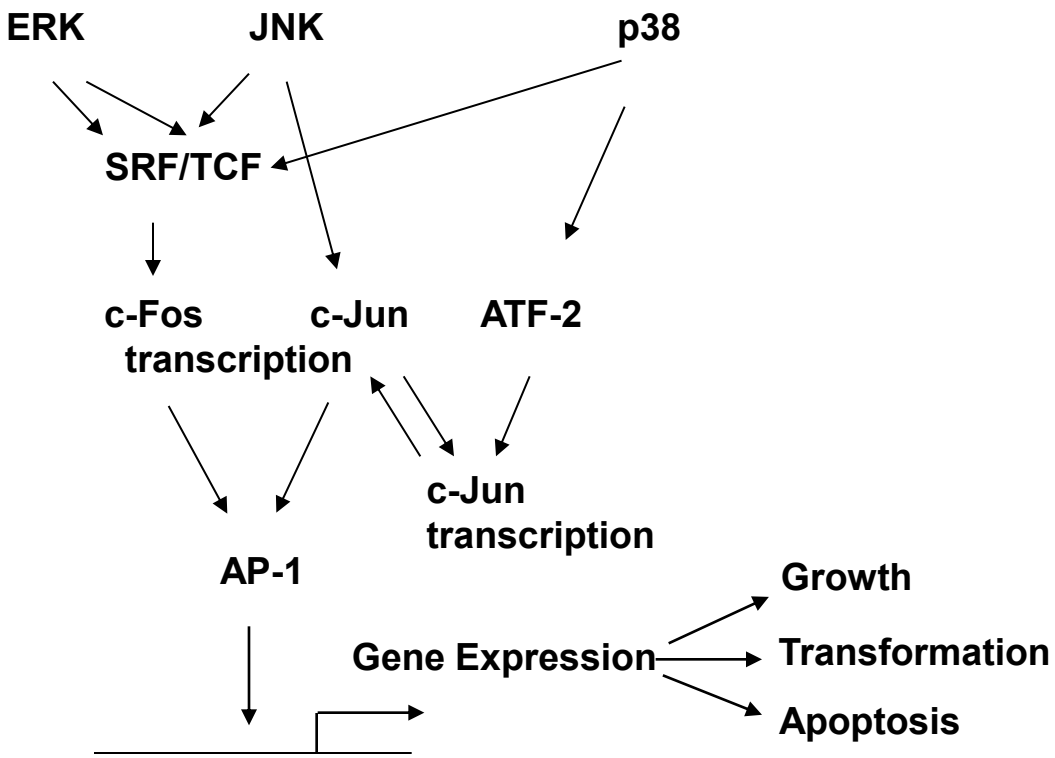
➢ The IKK complex is the convergent point of many extracellular stimuli that activate NF-κB



Regulation of Signal Transduction (cont'd)

✓ **Signaling convergency**

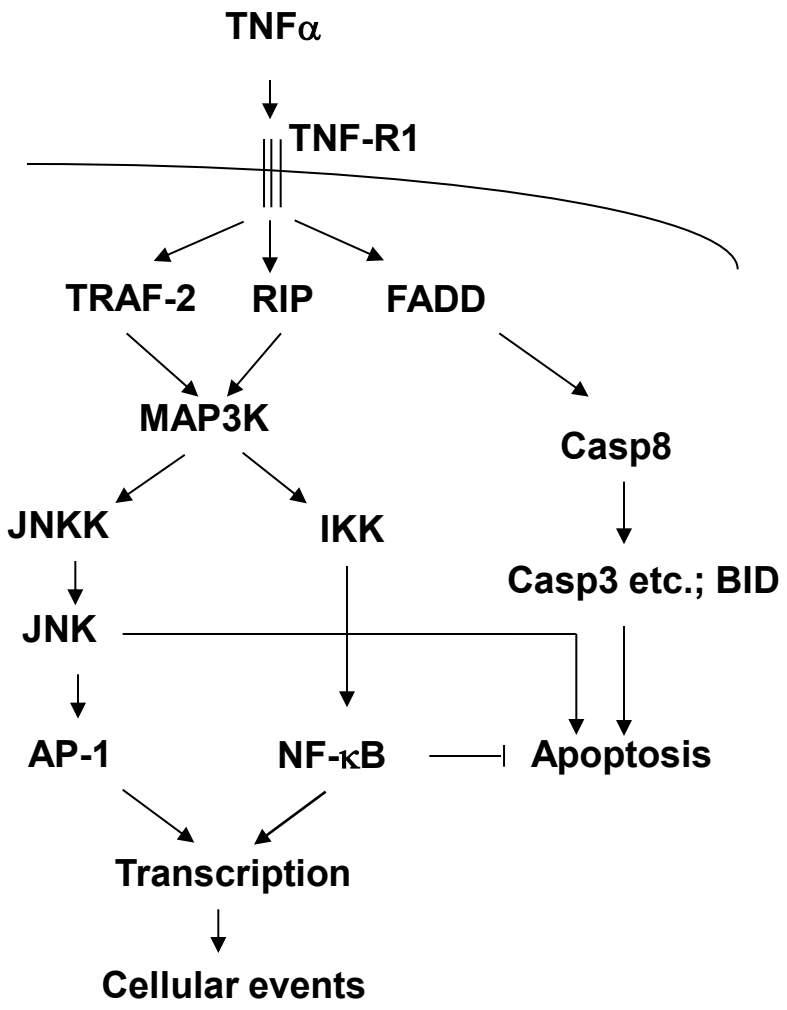
➢ **The AP-1 and SRF/TCF are convergent points for MAPK pathways**



Regulation of Signal Transduction (cont'd)

✓ **Signaling divergence:**

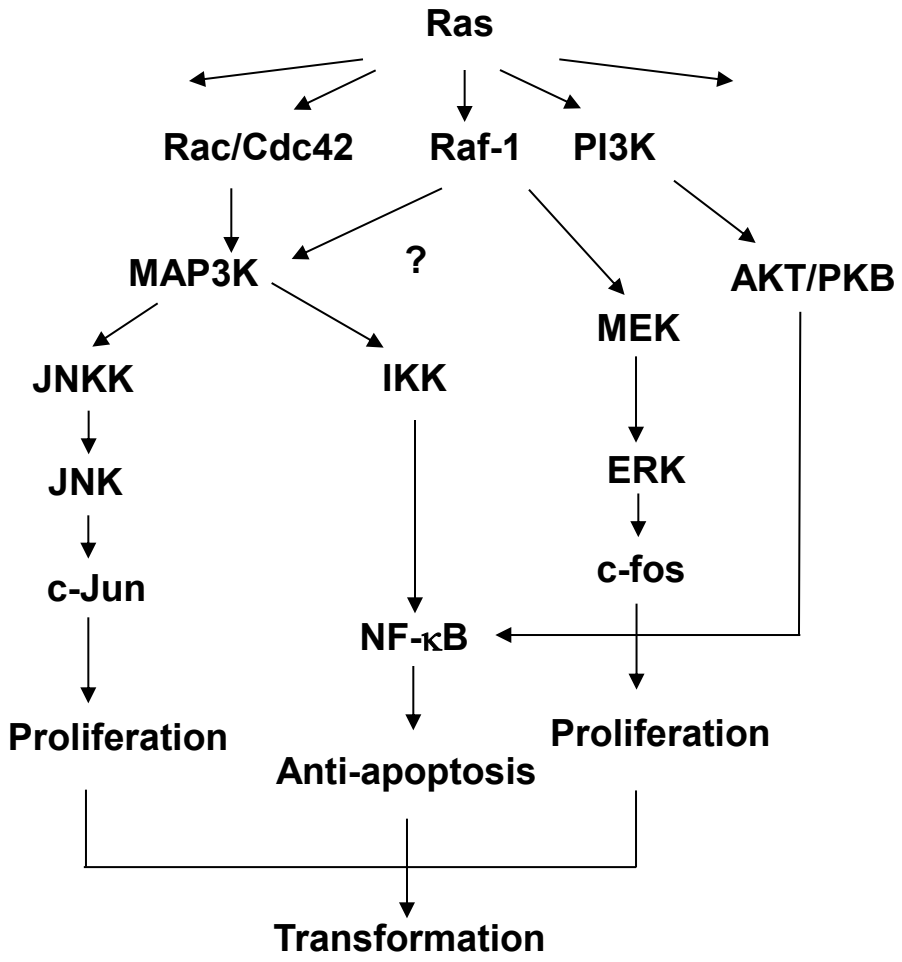
➢ **TNFα and its effector pathways**



Regulation of Signal Transduction (cont'd)

✓ **Signaling divergence:**

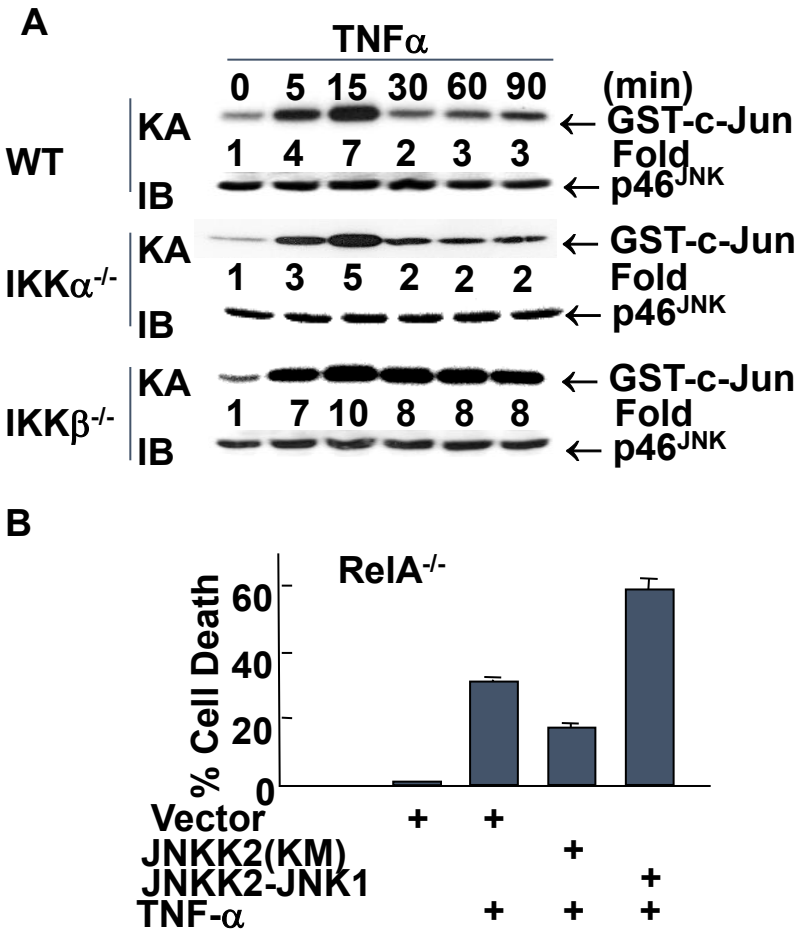
➢ **Ras and its effector pathways**

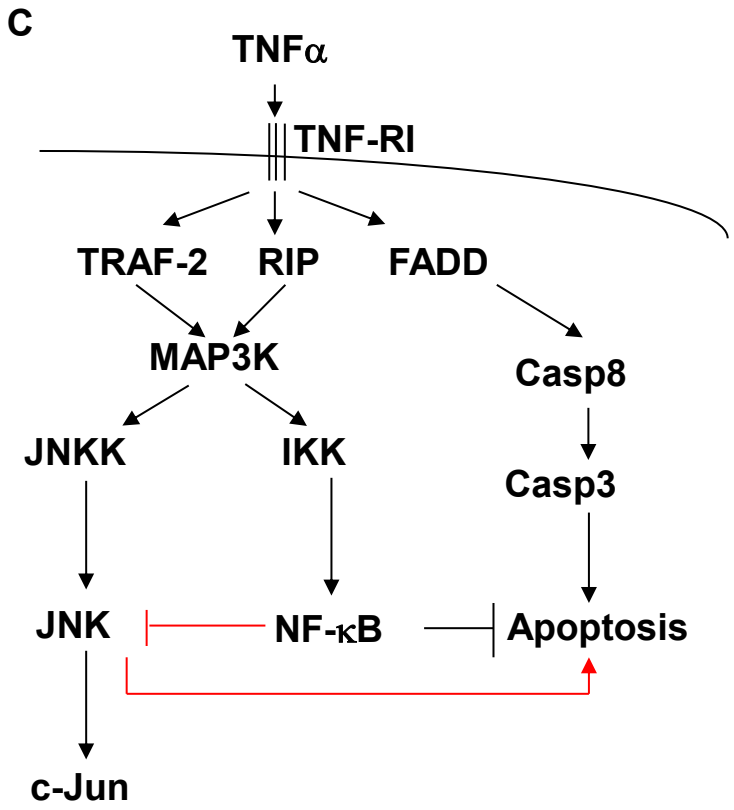


Regulation of Signal Transduction (cont'd)

✓ Signaling plasticity

➢ Negative regulation of JNK activation by NF-κB in response to TNFα



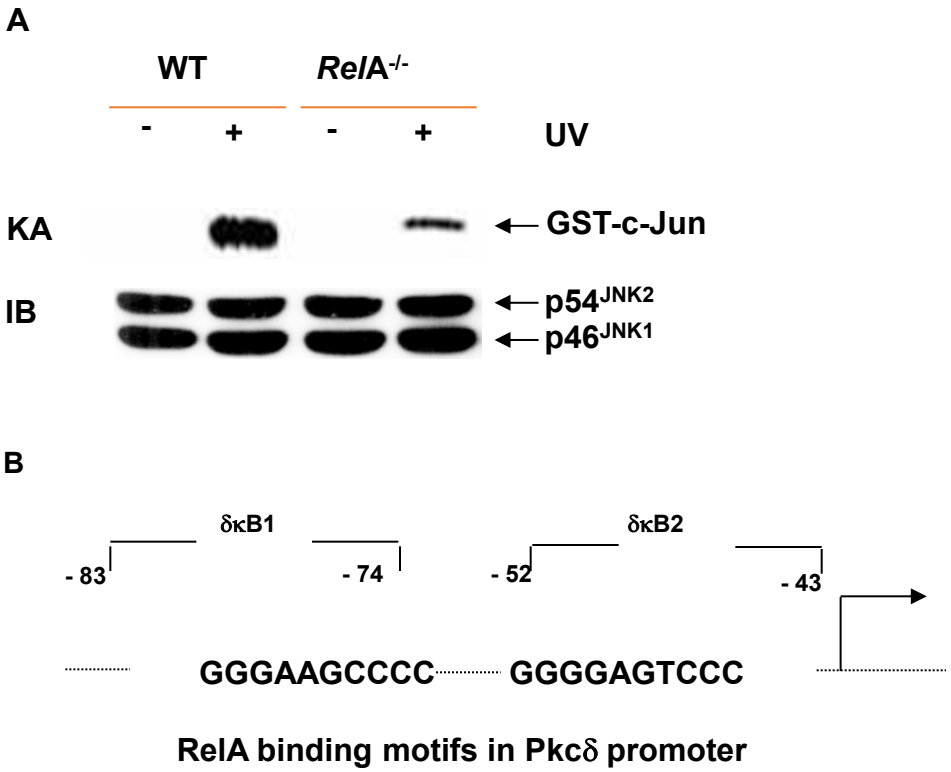


Tang *et al*., *Nature* 2001

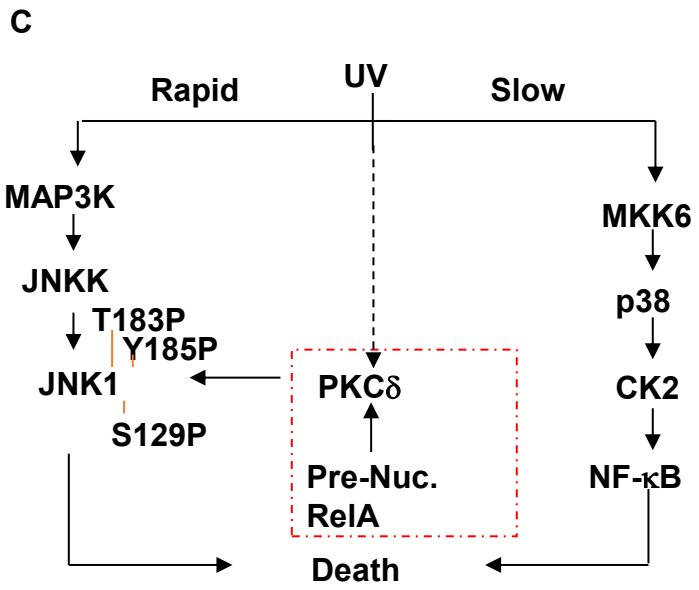
Regulation of Signal Transduction (cont'd)

✓ **Signaling plasticity (cont'd)**

➢ **Positive regulation of JNK activation by NF-κB in response to UV**



RelA binding motifs in Pkcδ promoter



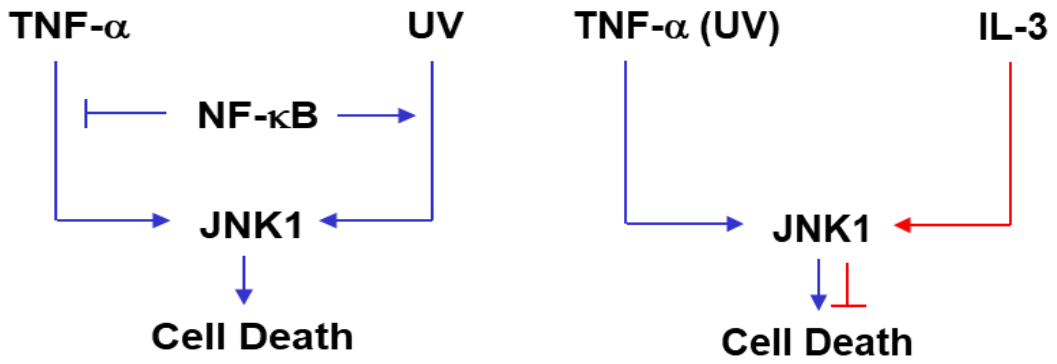
Liu *et al., Mol Cell* 2006

Regulation of Signal Transduction (cont'd)

✓ Signaling plasticity (cont'd)

➢ Some principles of signaling plasticity

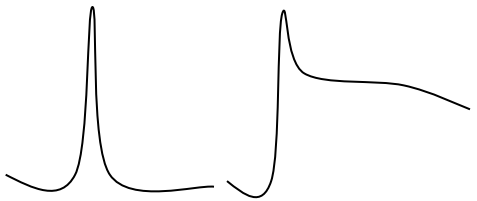
- ☐ The crosstalk between two given signaling pathways can have opposite biological outcomes in response to different stimuli;
- ☐ The same signaling molecule can have opposite biological functions in response to different stimuli.



Regulation of Signal Transduction (cont'd)

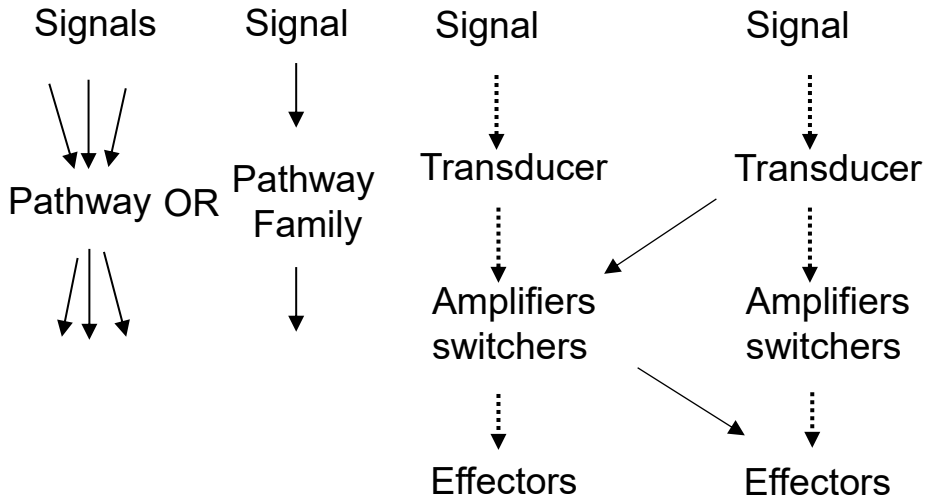
✓Pattern and threshold of signal transduction

- **Transient versus Sustained**



- Phos'n/dephos'n
- Degradation
- Inhibitors/feedback

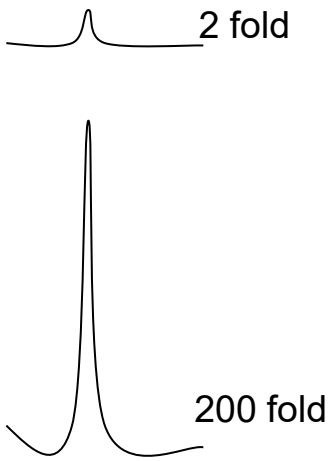
- **Redundancy versus Cross-talk**



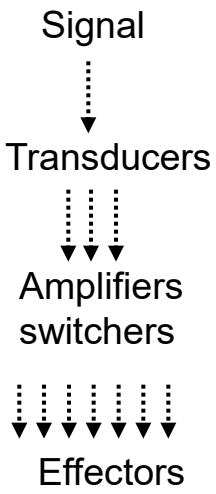
Regulation of Signal Transduction (cont'd)

✓Pattern and threshold of signal transduction (cont'd)

➢ Threshold versus Amplification



How many fold activation is enough?



Number + Enzymatic Rx

Common methods used in signaling study

Biochemistry and Molecular Biology Analysis

✓**Protein purification**

- FPLC
- Affinity

✓**Cloning:** Before and after the genomic project

- cDNA library screen
  - Oligonucleotide (EST+PCR)
  - In vitro hybridization
- PCR it!

✓**Subcloning**

- Expression vector construction
- Fusion protein: Tag-construction

Common methods used in signaling study

Biochemistry and Molecular Biology Analysis (cont'd)

✓ Phosphorylation and Dephosphorylation

- *In vitro* kinase (or phosphatase) assays
  - In vivo phosphorylation labeling: Really "hot"!
  - Two-dimensional phospho-peptide electrophoresis and amino acid analysis
- Site-directed mutagenesis
- Mass spectrometry and phospho-proteomics

✓ Protein-protein interaction

- Co-immunoprecipitation
- GST, His-tag pull-down assay
- Yeast two-hybridization

✓ Cell imaging

- Immunofluorescence staining
- Immunohistochemistry staining

Common methods used in signaling study (cont'd)

- ✓ **Transcription analysis**
  - Sequencing analysis
  - Electronic mobility shift assay (EMSA or band-shift assay)
  - Report gene assay
  - scRNA-seq
  - Single-cell transcriptome analysis
- ✓ **Genetic analysis**
  - Genetic screen for suppressors/enhancers/regulators
  - Gene edition (CRISPR/Cas)
  - Knock-out and knock-in
    - Cell type-specific: Alb-Cre, Vil-Cre
    - Constitutive versus Inducible: Cre/Loxp; Cre-ERT2
  - Transgenic
- ✓ **Specific activation and inhibition**
  - Constitutively active or dominant negative mutants
  - Specific chemical inhibitors: PD98059, SB202190, LY924002

Common methods used in signaling study (cont'd)

- ✓ **Specific activation and inhibition:**
  - Constitutively active or dominant negative mutants
  - Specific chemical inhibitors: PD98059, SB202190, LY924002
- ✓ **Disease models**
  - Genetic diseases
  - Patient specimen
- ✓ **AI Biology** – the future of biology